Jelena Tešić, Ph.D.

Associate Professor, Computer Science•Texas State University•San Marcos TX 78666 U.S.A. •jtesic@txstate.edu

EDUCATION

Ph.D. in Electrical and Computer Engineering, University of California Santa Barbara 01/2000 – 05/2004
Thesis title: Managing Large Multimedia Repositories
M.Sc. in Electrical and Computer Engineering, University of California, Santa Barbara, CA, 09/1998-12/1999
Outstanding Teaching Assistant Award, ECE department, June 1999.
Dipl. Ing. Electrical Engineering, University of Belgrade, Serbia, 10/1993-07/1998
Thesis title: “Noise reduction in CDMA receivers”, GPA: 9.64/10.00, top 1%.

PROFESSIONAL EXPERIENCE

2024- , Associate Professor; 2017-2024, Assistant Professor, Computer Science, Texas State University, TX.
2009-2017, Research Scientist, Mayachitra Inc, Santa Barbara, CA.
2004-2009, Research Staff Member, IBM Watson Research, NY.

RESEARCH FUNDING

- Feng Wang (PI), Jelena Tešić (co-PI), “Artificial Intelligence for Pavement Condition Assessment from 2D/3D Surface Images,” Texas Department of Transportation Phase 2 project, June 2026 – May 2028, $499,920.
- Mylene Farias (PI), **Jelena Tešić (co-PI),** Sarah Fritts (Supporting), “Advancing the Automation of Bat Monitoring for Sustainable Wind Energy Development, TXST Center for Analytics and Data Science, Texas State University, $6,000, 2025, Catalyst Award, grant.
- Xiangping Liu (PI), **Jelena Tešić (co-PI),** “Comprehensive Experiential Learning to Promote Climate Resilient Agriculture for Under-Represented Minority Students,” USDA NIFA HSI, $1,200,000, Oct 2024 – Sep 2028, grant.
- Denise Gobert (PI), Monica Hughes (Co-PI), **Jelena Tešić (co-PI),** Keri Jackson, Keri (Supporting), “Hospital Readmissions for Persons with Diabetes who Receive Home Health Services: Predictive Model Analysis using Machine Learning,” Williamson County Foundation Grant, Private / Foundation / Corporate, $10,000, Nov 2024 – Aug 2026, grant.
- Tahir Ekin, Tahir (Pl), **Jelena Tešić (co-PI),** Apan Qasem (co-PI), Damian Valles Molina (co-PI), Lucia Summers (co-PI), “Expanding AI Curriculum and Infrastructure at Texas State University to Advance Interdisciplinary Research and Grow a Diverse AI Workforce, NSF Expand AI, Federal, $400,000, Jan 2024 – Dec 2025, 400K, grant.
- Robert McLean (PI) and **Jelena Tešić (co-PI),** “Automating Microbial Biofilm Corrosion Analysis Using Semantic Segmentation for Enhanced Space Mission Safety, Texas State University, $16,000, 2024, grant.
- Feng Wang, Feng (PI), Haitao Gong (Co-PI), **Jelena Tešić (Co-PI),** Xiaohua Luo, Xiaohua (Co-PI), “Artificial Intelligence for Pavement Condition Assessment from 2D/3D Surface Images, Texas Department of Transportation (TxDOT),” State, $451,875, Sep 2022- Aug 2025, grant.
- Salah Faroughi (PI), **Jelena Tešić (co-PI),** John Tiefenbacher, (Co-PI), “ESMs Latent Space Exploration for Uncertainty Quantification and Spatiotemporal Downscaling,” DOE-BER, $150,000. Aug 2022 – Jun 2025, grant.
- **Jelena Tešić (PI)**, “Object Cueing Using Biomimetic Approaches to Visual Information Processing,” NAVAIR, Federal, NAVAIR SBIR N14A-T008 AT, $199,708, March 2021 – March 2023, grant @ TXST; NAVAIR SBIR N14A-T008 P2, $307,005, March 19, 2018 – March 2021, grant @ TXST; NAVAIR STTR N14A-T008, P1 and P2, $1,200,000 Oct 2014 – Jan 2018, grant a@ Mayachitra Inc.

PROFESSIONAL ACTIVITIES

7 U.S. PATENTS AWARDED: 12,093,970; 9,710,760; 8,738,695; 8,032,539; 7,958,068; 7,818,329; 7,707,162.
Organization ACM ICMR 2026 proceedings chair, ACM Web 2027 Tutorial chair, ACM ICMR 2027 TPC, and ACM MMSys 2028 organizing committee.
Guest Editor Special Issue on Collaborative Tagging of Multimedia, IEEE Multimedia Magazine.

Panelist: NSF CISE/IIS III CRII and SMALL Programs; ACMMM "Diversity in Multimedia Retrieval Research"
Area Chair, Section Host, and Reviewer for numerous ACM and IEEE conferences, and reviewer for numerous ACM, IEEE, Elsevier, Springer, and Cambridge Journals.
Founder, DataLab@TXST (https://DataLab12.github.io), 2018 - present.
Co-Founder, Research Lead, Texas State University Center for Analytics and Data Science (CADS), 2023 – present.
IEEE and National Academy of Inventors (NAI) Senior Member

PUBLICATIONS

Full publication list: https://scholar.google.com/citations?hl=en&user=jRLy9uoAAAAJ, ***h-index: 24, i-10 index: 43.***

PUBLICATIONS @ TXST - *TXST Student authors in italic*; IF: Impact Factor; IS: Impact Score.

1. *MMM Rahman*, Y. Lu, **J. Tešić,** (2026). COAST: Cross-dataset Omics And Spatial Transcriptomics Atlas via Multi-Modal Approximate Nearest Neighbor Search. SIAM International Conference on Data Mining (SDM26), Oct 2026, h5-index: 36 and h5-median: 66.
2. *T. Tani*, TA Busboom, RJC McLean, and **J. Tešić**, (2026). Benchmarking Automated MIC Detection: The AI-Ready Space Biology SEM Dataset and Advanced Detection Methods. npj Microgravity, to appear.
3. *M. Elizondo, CM Jones*, **J. Tešić (2026)**, "N3C Data Analytics for Attribute Importance and Prediction Tasks," Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (BCB '26). Poster. https://doi.org/10.1145/3807503.3817110, h5-index: 28; h5-median: 50.
4. *M Elizondo*, DJ Amante, **J Tešić (2026), "**PRECISION-Connect: AI-Ready Multimorbidity and SDOH Risk Vectors for Explainable 30-Day Readmission and County-Level Disparity Modeling," Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (BCB'26), Short paper. https://doi.org/10.1145/3807503.3819383, h5-index: 28; h5-median: 50.
5. *M. Shebaro*, **J. Tešić** (2026), Scaling frustration index and balanced state discovery for real signed graphs, Journal of Computational Science, Volume 100, No. 102981, ISSN 1877-7503, https://doi.org/10.1016/j.jocs.2026.102981, IF:4.0, CiteScore: 8.8, h-index: 78.
6. *W. Tao*, Y. Bai, F. Wang, and **J. Tešić** (2026), "Data-Centric Multimodal Pavement Distress Detection Using Intensity–Range Imagery. In Proceedings of the 2026 International Conference on Multimedia Retrieval (ICMR '26). Full paper. https://doi.org/10.1145/3805622.381061, h5-index: 33.
7. *E Zeraatkar*, **J Tešić** (2026), "SAFIR: Spatially-aware Activation Function for Implicit Neural Representations," Computer Vision and Image Understanding, Volume 268, No. 104754, ISSN 1077-3142, https://doi.org/10.1016/j.cviu.2026.104754, JIF: 3.6, 5-year IF: 4.3, CiteScore: 7.2.
8. *D. Biswas, A. Scouten,* H. Gong, F. Wang, and **J. Tešić (2026),** "MoPac+: Multimodal Modeling of the Pavement Cracks using Hard-Example Mining and Context-Aware Feature Aggregation," MEAS, 2026, IF 5.6.
9. *E. Zeraatkar,* S. Faroughi, and J**. Tešić (2026)**, "Frequency-Aware Vision Transformers for High-Fidelity Super Resolution of Earth System Models," Nature Scientific Reports, Springer, 2026, IF 3.9.
10. *D Biswas*, **J Tešić** (2025), "MMVAD: A Vision-Language Model for Cross-Domain Video Anomaly Detection with Contrastive Learning and Scale-Adaptive Frame Segmentation," Elsevier Expert Systems with Applications Journal (IF 7.5)
11. *MMM* Rahman, Jelena Tešić (2025), "Accelerating Vector Search at Scale: BAM-ANN with Batch-Aware Memory-Disk Hybrid Indexing," 2025 IEEE International Conference on Content-Based Multimedia Indexing (CBMI), Dublin, Ireland, h-index: 22.
12. *M Shebaro*, L Feng, **J Tešić** (2025), ABCD: Algorithm for Balanced Component Discovery in Signed Networks, IEEE Access (IF: 3.6)
13. *M Elizondo, J Yu, D Payan*, L Feng, **J Tešić** (2025), Novel Considerations in the ML/AI Modeling of Large-Scale Learning Loss, IEEE Access (IF: 3.6)
14. *T Tani, A Scouten, E Ortiz*, R McLean, **J Tešić** (2025), "Automated Corrosion Identification in Metal Imagery: Traditional vs. Deep Learning,", Advances in Visual Computing. Lecture Notes in Computer Science, vol 15047. Springer, Cham.
15. *H Gong*, A Scouten, **J Tešić**, F Wang (2025) "Deep Learning Pipeline for Modeling Pavement Cracks with an Imbalanced Dataset (MoPaC)", TRR: Journal of the Transportation Research Board (IF: 2.1)
16. *D Biswas*, **J Tešić** (2024), "Domain Adaptation with Contrastive Learning for Object Detection in Satellite Imagery", IEEE Transactions on Geoscience and Remote Sensing (IF:8.2, IS:10.85).
17. *M Elizondo*, D Gobert, **J Tešić** (2024), "AI/ML Pipeline for Predicting Diabetic Readmissions from CMS OASIS dataset", IEEE Big Data Poster (IF:7.2).

18. *MMM Rahman*, **J Tešić** (2024), "Stratified Graph Indexing for Efficient Search in Deep Descriptor Databases" Springer International Journal of Multimedia Information Retrieval (IF:5.6).
19. *D Biswas*, **J Tešić** (2024) "Unsupervised Domain Adaptation with Debiased Contrastive Learning and Support-Set Guided Pseudo Labeling for Remote Sensing Images: JSTARS-2023-01182, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (IF: 5.5, IS:5.5)
20. *M. Shebaro, L Nogueira De Moira*, **J Tešić** (2024) "Improving Association Discovery through Multiview Analysis of Social Networks", Springer Nature Social Network Analysis and Mining (IF: 2.8, IS:3.5).
21. *T.A. Tani*, **J Tešić** (2024) "Advancing Retinal Vessel Segmentation with Diversified Deep Convolutional Neural Networks, IEEE Access (IF:3.5).
22. *MMM Rahman*, *D Biswas,* **J Tešić** (2024) "Image Deduplication Using Efficient Visual Indexing and Retrieval: Optimizing Storage, Time and Energy for Deep Neural Network Training," submitted to Springer Signal Image and Video Processing Journal (IF 2.3).
23. *D. Biswas*, **J Tešić** (2024) *"BinaryDNet53: A Lightweight Binarized CNN for Monkeypox Virus Image Classification*," in Springer Signal Image and Video Processing Journal (IF: 2.3).
24. *M. Shebaro*, **J Tešić** (2023) "Identifying Stable States of Large Signed Graphs," WWW '23 Companion, ATX (IS:15.6).
25. *H Gong*, **J Tešić**, *J Tao*, X Luo, F Wang (2023) "Automated Pavement Crack Detection with Deep Learning Methods: What Are the Main Factors and How to Improve the Performance?", Transportation Research Record: Journal of the Transportation Research Board (IF: 2.1)
26. *M Elizondo*, R Musal, *J Yu*, **J Tešić** (2013) "Long COVID Challenge: Predictive Modeling of Noisy Clinical Tabular Data," IEEE ICHI, June 2023, Houston, TX (IS:0.6, research ranking medicine:55).
27. *D Biswas*, **J Tešić** (2022) "Small Object Difficulty (SOD) Modeling for Objects Detection in Satellite Images", In 14th IEEE International Conference on Computational Intelligence and Communication Networks (CICN), virtual.
28. *MMM Rahman*, **J Tešić** (2022), "Evaluating Deep Feature Approximate Nearest Neighbor Indexing and Search," 2022 IEEE International Conference on Big Data (Big Data), virtual (IS:4.10).
29. *MMM Rahman*, **J Tešić** (2022), "Hybrid Approximate Nearest Neighbor Indexing and Search (HANNIS) for Large Descriptor Databases," 2022 IEEE International Conference on Big Data (Big Data), virtual (IS:4.10).
30. *D Biswas, T Rahman*, Z Zong, **J Tešić** (2022) "Improving the Energy Efficiency of Real-time DNN Object Detection via Compression, Transfer Learning, and Scale Prediction", IEEE NAS, Oct 2022, Philadelphia, PA (IS:10.65).
31. A Lommatzsch, B Kille, Ö Özgöbek, *Y Zhou*, **J Tešić** et al. (2022) NewsImages: addressing the depiction gap with an online news dataset for text-image rematching, ACM MM Sys, June 2022, Athleone, Ireland (IS:4.15).
32. *M. Tomasso*, L Rusnak, **J Tešić** (2022). Advances in Scaling Community Discovery Methods for Large Signed Graph Networks, *Oxford Journal of Complex Networks* Vol. 10, Issue 3, (IF:2.1, IS:2.18).
33. *M Tomasso*, L Rusnak, **J Tešić** (2022). Cluster Boosting and Data Discovery in Social Networks, Proceedings of the ACM Symposium on Applied Computing (SAC), (IS:1.33).
34. *L Nogueira De Moira*, **J Tešić** (2021). pytwanalysis: Twitter Data Management and Analysis at Scale, IEEE International Conference on Social Networks Analysis, Management and Security (SNAMS), (IS:0.6).
35. *G Strauch, JJ Lin*, **J Tešić** (2021). Overhead Projection Approach for Multi-Camera Vessel Activity Recognition, online IEEE Big Data, REU Symposium track (IS:3.64).
36. *G Alabandi*, **J Tešić**, L Rusnak, M Burtscher (2021). Discovering and balancing fundamental cycles in large, signed graphs, Proceedings of the International Conference for High-Performance Computing (IS:5.2).
37. L Rusnak, **J Tešić** (2021). Characterizing Attitudinal Network Graphs through Frustration Cloud. Springer *Data Mining and Knowledge Discovery*, 35, 2498–2539 (IF:4.8; IS:5.6).
38. *B Ford*, A Qasem, **J Tešić**, Z Zong (2021) "Migrating Software from x86 to ARM Architecture: An Instruction Prediction Approach", IEEE NAS (IS:11.8).
39. *A. Magill, LN De Moura, M. Tomasso, M. Elizondo*, and **J. Tešić** (2020) "Enriching content analysis of tweets using community discovery graph analysis," Proceedings of the MediaEval 2020 Workshop, volume 2882.
40. **J Tešić**, D Tamir, S Neumann, N Rishe, A Kandel, "Computing with Words in Maritime Piracy and Attack Detection Systems," International Conference on Human-Computer Interaction, 434-444 (IS:4.11).
41. *N Dunstatter, A Tahsini*, M Guirguis, **J Tešić**, "Solving Cyber Alert Allocation Markov Games with Deep Reinforcement Learning," International Conference on Decision and Game Theory for Security, 164-183 (IS:3.3).
42. *H Samimi*, **J Tešić**, AHH Ngu, "Patient Centric Data Integration for Improved Diagnosis and Risk Prediction," Heterogeneous Data Management, Polystores, and Analytics for Healthcare, 185-195 (IS:3.22).

43. *DB Heyse, N Warren*, **J Tešić**, “Identifying maritime vessels at multiple levels of descriptions using deep features,” SPIE Artificial Intelligence and Machine Learning for Multi-Domain Operations (IS:0.49).
44. *T Mauldin*, AH Ngu, V Metsis, ME Canby, **J Tešić**, Experimentation and analysis of ensemble deep learning in IoT applications, Open Journal of Internet of Things (OJIOT) 5 (1), 133-149, 2019 (IF:2.38).
45. *N Warren, B Garrard*, E Staudt, **J Tešić**, Transfer learning of deep neural networks for visual collaborative maritime asset identification, IEEE 4th International Conference on Collaboration and Internet, 2018 (IS:0.76).

EXTENDED ABSTRACTS

1. *M. Shebaro*, **J Tešić** (2023) “Identifying Stable States of Large Signed Graphs,” Sunbelt International Network for Social Network Analysis (IS:0.6).
2. *A Pantoja, E Bilewu*, **J Tešić** (2021) “Deep Learning Modeling and Vehicle Tracking for Smart City Traffic Alert System,” Bulletin of the American Physical Society (IS:1.6).
3. *L Nogueira De Moira*, **J Tešić** (2019) “Spread of English Neologisms through Brazilian Portuguese Online Chatter, A Data Science Perspective,” The 8th Hispanic and Luso-Brazilian Linguistics Conference at Arizona State University.